



PRODUCT NAME

Heavy-duty Powder Detergent Formula

Description

laundry powder detergent

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Carbonate	Sodium Carbonate	27.75%	pH buffering agent and builder
2	Sodium Sulfate	Sodium Sulfate	16.65%	filler to enhance bulk and reduce cost
3	Sodium Tripolyphosphate	Sodium Tripolyphosphate	16.65%	builder to enhance cleaning performance
4	Surfactant Blend (Anionic/Cationic)	Fatty Alcohol Ethoxylate, Alkylbenzene Sulfonic Acid	22.2%	primary cleaning agent
5	Sodium Bicarbonate	Sodium Bicarbonate	5.55%	mild alkali to support pH balance and stain removal
6	Enzymes (Protease, Amylase)	Protease, Amylase	3.33%	stain removal through biological action
7	Polymer (Polyacrylate)	Polyacrylate	2.22%	anti-caking agent and dispersant
8	Fragrance	Fragrance	1.11%	provide pleasing scent to the finished product
9	Colorant	Colorant	0.11%	aesthetic enhancement
10	Water (for granulation only)	Water	4.43%	moisture for processing (minimum for granulation)
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions